ELECTRICAL MACHINES · CH7

Induction machine fundamentals

Cheat sheet · Day 10 · Mon 2 Nov · pack M7

Reference-sheet lines

COPY THIS ONTO YOUR PRINTED REFERENCE SHEET

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INDUCTION MACHINE FUNDAMENTALS (Ch7)
n_S = 60f/p (p = pole PAIRS) = 120f/poles     w_S = 2*pi*n_S/60
s = (n_S - n)/n_S      n = (1-s) n_S      f2 = s f      slip speed = n_S - n
s=1 standstill (f2=f), s=0 sync (R2'/s = inf), s<0 generator (R2'/s < 0)
Keep s to 4 s.f.

SIMPLIFIED CCT (per phase):  V1 || (RC // jXM)  then  R1 + jX + R2'/s,  X = X1 + X2'
  star: V1 = VL/sqrt3, I1 = IL     delta: V1 = VL, IL = sqrt3 I1     V1 at 0 deg
  I2' = V1/(R1 + R2'/s + jX)    IM = V1/RC - jV1/XM    I1 = I2' + IM  (rectangular!)
R2'/s = R2' (heat) + R2'(1-s)/s (mech power)

POWER FLOW (3-ph totals)
  P_IN = 3 V1 I1 cos(phi)
  - stator Cu 3 I2'^2 R1   - iron 3 V1^2/RC
  = P_AG = 3 I2'^2 R2'/s = τ w_S
  rotor Cu = 3 I2'^2 R2' = s P_AG      P_M = (1-s) P_AG = τ w
  ideal eta <= 1 - s = n/n_S          eta = P_M/P_IN

TORQUE   τ = P_AG/w_S = P_M/w = 3 I2'^2 R2'/(s w_S)
  τ = 3 V1^2 (R2'/s) / { w_S [ (R1 + R2'/s)^2 + (X1+X2')^2 ] }    (V1 = PHASE V)
  τ ~ V1^2 at fixed s;  low s: τ ~ s;  high s: τ ~ 1/s

TYPE C STEPS: V1 -> n_S, s, R2'/s -> draw cct -> I2' -> IM -> I1, pf -> P_IN
              -> P_AG -> τ = P_AG/w_S (check formula) -> P_M -> eta (check loss sum)
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